CNS Lab Experiment-1

Procedure

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Objective: To implement **user authentication techniques** for **remote access** of network devices (Router and Switch) using **local username–password authentication** and **SSH** in Cisco Packet Tracer.

Step 1: Network Topology Setup

Take a router, a switch and a PC; interconnect using straight-through cables to form the required network topology. Then assign IP Address to PC:

IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

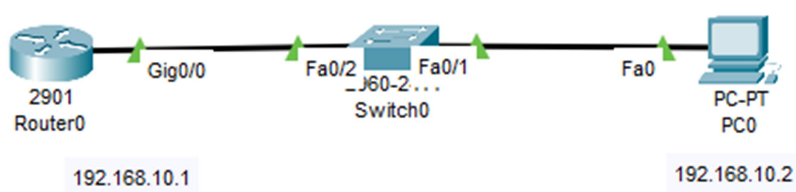

Fig.1. Topology

Part A: Router Configuration: Local User Authentication (Telnet)

Step 2: Enter Router CLI and Configure Router Interface

```
Router> enable
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 3: Creation of Local User Account

```
Router(config)# username CNSLAB privilege 15 password cnslab
```

This command creates a local user account in the router's local database. It enables user-based authentication instead of a single shared password. It allows **controlled access** to the network device that helps in accountability and auditing, as each user logs in with a unique username. Forms the foundation for secure remote access using Telnet or SSH. Here, *CNSLAB* is the **username** assigned to the user, *privilege 15* grants the user **full administrative access**, *password cnslab* sets the password for user authentication.

Step 4: Enabling Local Authentication on VTY Lines

VTY lines (Virtual Teletype lines) are logical interfaces on a Cisco network device that allow remote access to the device over the network using Telnet or SSH.

- VTY lines are not physical ports; they are virtual channels.
- They are used when an administrator connects to a router or switch from a remote PC.
- Each VTY line supports one remote session at a time.
- Commonly, Cisco devices have VTY lines 0 to 4 (five simultaneous remote users), though some devices support more.

CLI Commands:

```
Router(config)# line vty 0 4
Router(config-line)# login local
Router(config-line)# transport input telnet
```

Enabling Local Authentication on VTY Lines ensures that only authenticated users can access the router remotely that prevents unauthorized login attempts. It links the VTY lines directly to the local username–password mechanism that allows controlled remote administration of the device. Here,

- `line vty 0 4` selects the virtual terminal lines used for remote access.
- `login local` instructs the router to authenticate users using the local user database.
- `transport input telnet` allows Telnet-based remote connections.

Step 5: Encrypting Plaintext Passwords

```
Router(config)# service password-encryption
```

This command encrypts all plaintext passwords stored in the router configuration.

Passwords are encrypted using Type 7 encryption.

Step 6: Setting Enable Secret Password

```
Router(config)# enable secret class
```

This command sets a secure encrypted password for entering privileged EXEC mode. `enable secret` uses MD5 hashing. It overrides any existing `enable password` command.

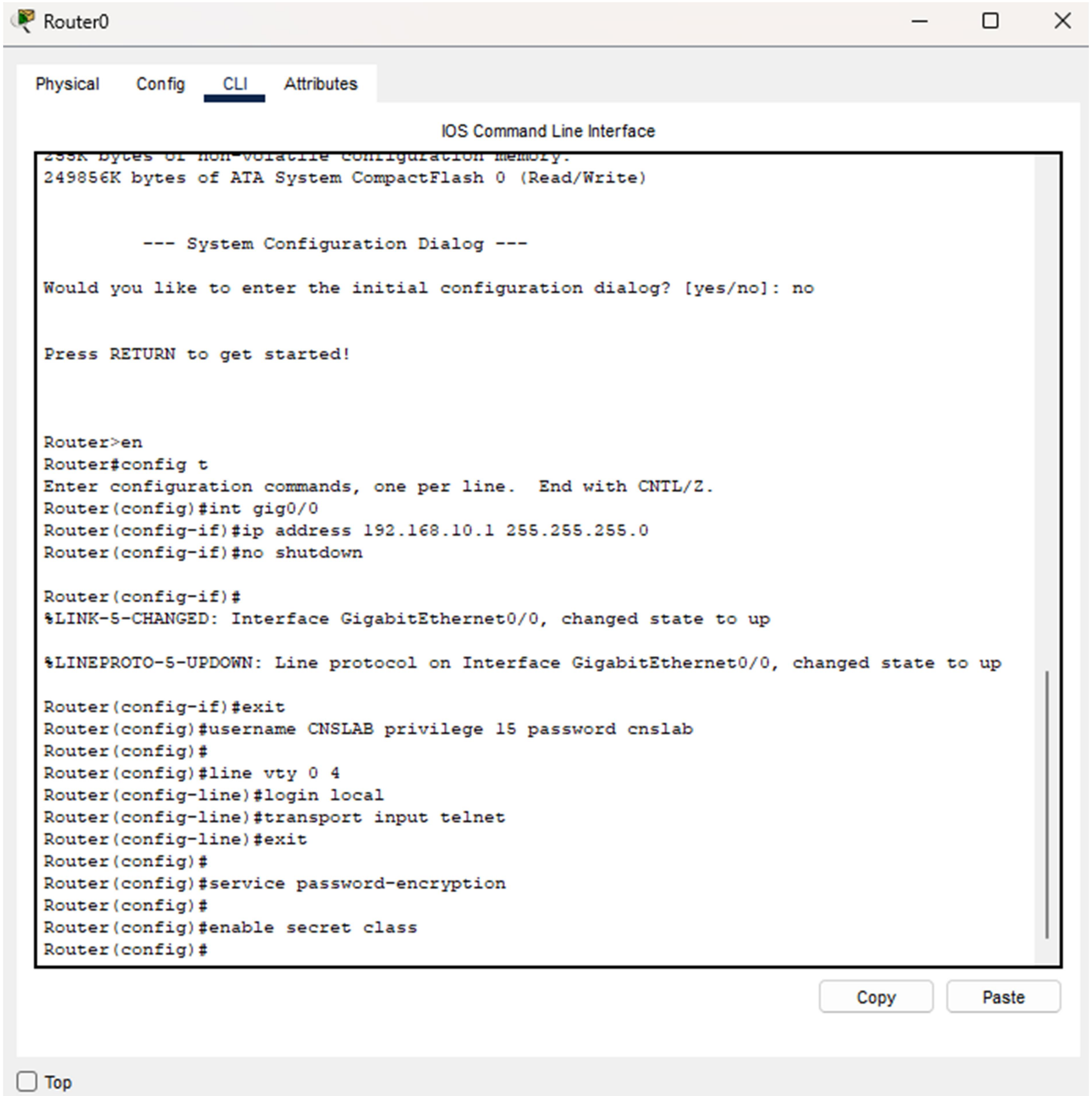

Fig.2. Up to Step 6

Step 7: Verification of Telnet Authentication

- Click PC → Desktop → Command Prompt
- Type: telnet 192.168.10.1
- Enter: Username: CNSLAB   Password: cnslab

Successful login confirms local user authentication using Telnet.

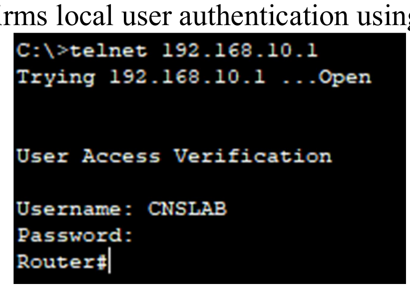

Fig.3. Authentication verification

Part B: Router Configuration: SSH Authentication

Step 8: Configure Hostname and Domain Name

```
Router(config)# hostname R1
R1(config)# ip domain-name cnslab.com
```

- **`hostname R1:`** assigns a unique identity to the router.
- **`ip domain-name cnslab.com:`** configures the domain name required for SSH.
- The hostname and domain name together form the fully qualified domain name (FQDN) of the device.
- Configuration of Hostname and Domain Name is mandatory for SSH configuration**.** It is used during RSA key generation and helps in device identification in large networks.

Step 9: Generate RSA Keys

```
R1(config)# crypto key generate rsa
```

- This command generates RSA public and private keys on the router.
- These keys are used for encryption and authentication in SSH.
- When prompted, a key size (e.g., 1024 bits) is specified.

Step 10: Enable SSH and Disable Telnet

```
R1(config)# ip ssh version 2
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit
```

- **`ip ssh version 2:`** enables SSH version 2, which is more secure than version1.
- **`line vty 0 4:`** selects remote access lines.
- **`transport input ssh:`** allows only SSH connections and blocks Telnet.
- **`login local:`** enforces local username–password authentication.

Overall Significance of Step 10

- It converts remote access from insecure Telnet to secure SSH.
- It protects network devices from eavesdropping and credential theft**.**

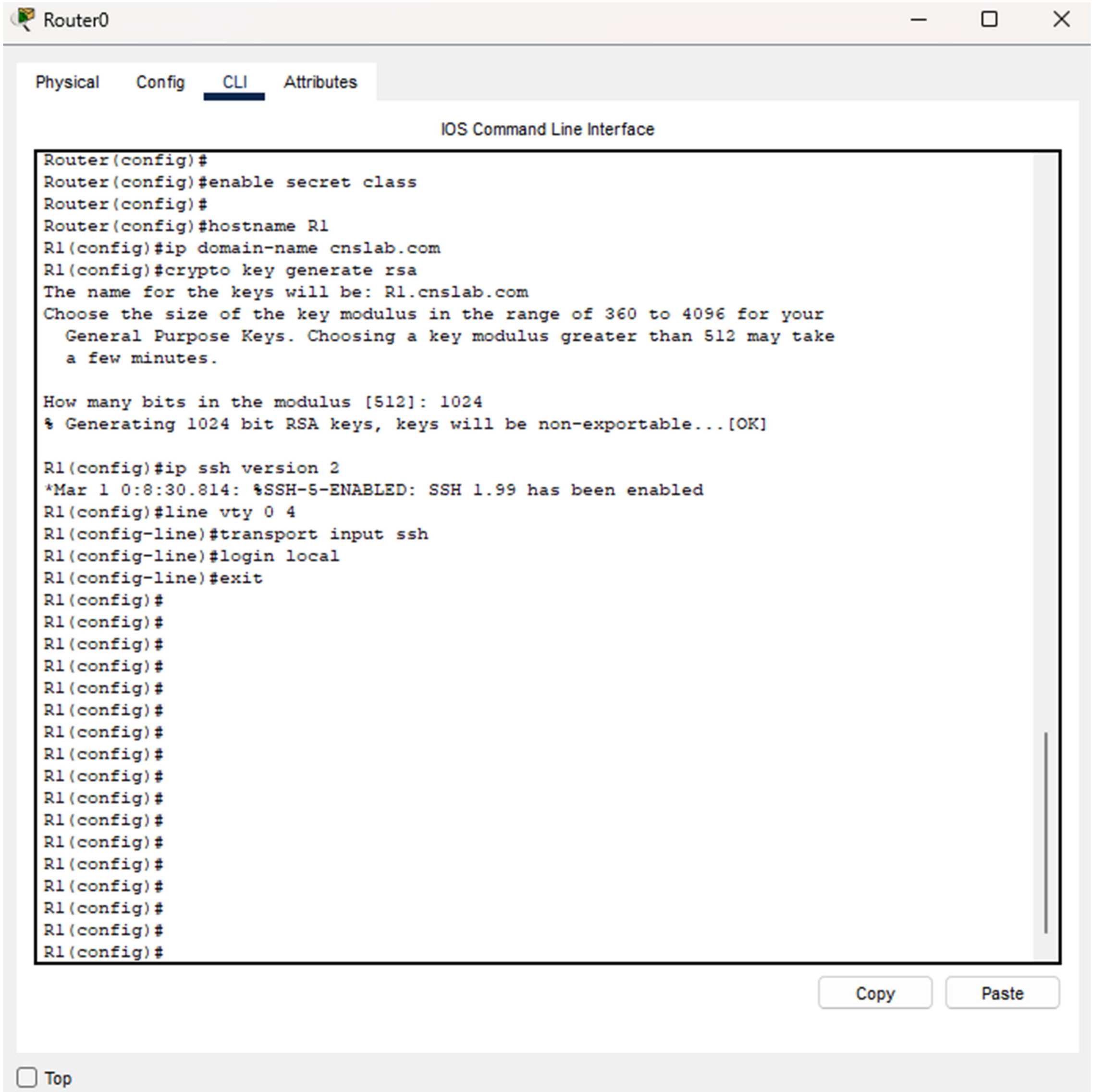

Fig.4. Steps 8, 9 and 10

Step 11: Verification of SSH Authentication

- Open PC → Desktop → Command Prompt
- Type: `ssh -l CNSLAB 192.168.1.1`
- Enter password: `cnslab`

Successful login confirms secure remote access using SSH.

Fig.5. SSH Authentication verification

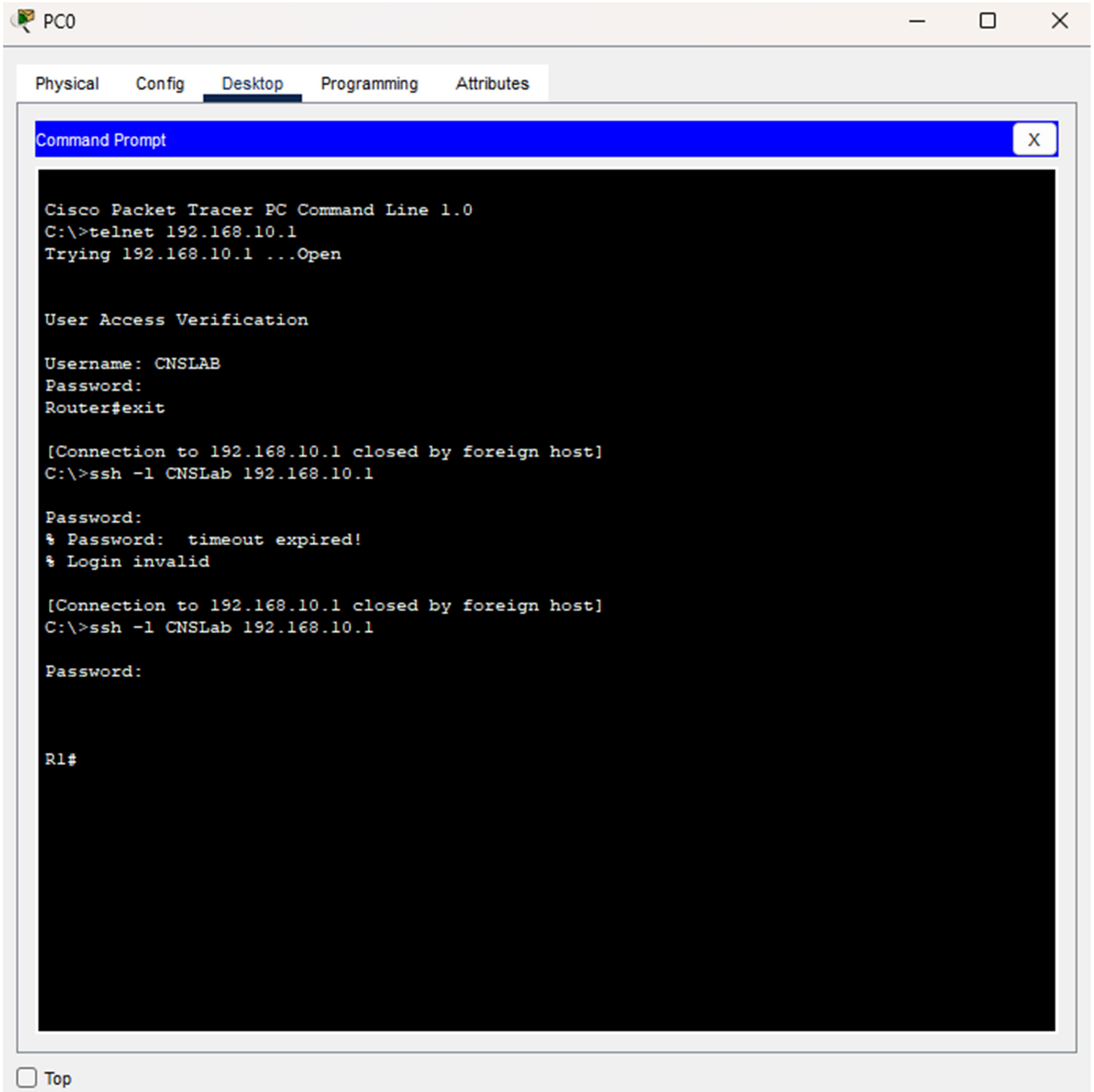

Fig.6. Both Telnet and SSH Authentication verification